



Taurine: The comeback of a nutraceutical in the prevention of retinal degenerations



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ABSTRACT

Taurine is the most abundant amino acid in the retina. In the 1970s, it was thought to be involved in retinal diseases with photoreceptor degeneration, because cats on a taurine-free diet presented photoreceptor loss. However, with the exception of its introduction into baby milk and parenteral nutrition, taurine has not yet been incorporated into any commercial treatment with the aim of slowing photoreceptor degeneration. Our recent discovery that taurine depletion is involved in the retinal toxicity of the antiepileptic drug vigabatrin has returned taurine to the limelight in the field of neuroprotection. However, although the retinal toxicity of vigabatrin principally involves a deleterious effect on photoreceptors, retinal ganglion cells (RGCs) are also affected. These findings led us to investigate the possible role of taurine depletion in retinal diseases with RGC degeneration, such as glaucoma and diabetic retinopathy. The major antioxidant properties of taurine may influence disease processes. In addition, the efficacy of taurine is dependent on its uptake into retinal cells, microvascular endothelial cells and the retinal pigment epithelium. Disturbances of retinal vascular perfusion in these retinal diseases may therefore affect the retinal uptake of taurine, resulting in local depletion. The low plasma taurine concentrations observed in diabetic patients may further enhance such local decreases in taurine concentration. We here review the evidence for a role of taurine in retinal ganglion cell survival and studies suggesting that this compound may be involved in the pathophysiology of glaucoma or diabetic retinopathy. Along with other antioxidant molecules, taurine should therefore be seriously reconsidered as a potential treatment for such retinal diseases.

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Abbreviations: CDO, cysteine dioxygenase; CNS, central nervous system; CSAD, cysteine sulfonic decarboxylase; DAPI, 4',6-diamidino-2-phenylindole; DIV, days *in vitro*; DM, diabetes mellitus; DR, diabetic retinopathy; ERG, electroretinogram; GABA, γ -amino butyric acid; GFAP, glial fibrillary acidic protein; GCL, ganglion cell layer; INL, inner nuclear layer; NMDA, N-methyl D-aspartate; ONL, outer nuclear layer; OPL, outer plexiform layer; PAT1, proton-coupled amino acid transporter; RGC(s), retinal ganglion cell(s); ROS, reactive oxygen species; SAH, S-adenosylhomocysteine; SAM, S-adenosyl methionine; Tau-T, taurine transporter; VGB, vigabatrin; RP, retinitis pigmentosa; RPE, retinal pigment epithelium; Tau-T KO, taurine-transporter knockout.

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1. Taurine: a “semi-essential” sulfur amino acid in the animal kingdom

1.1. Origin of taurine

2-Amino-ethanesulfonic acid, commonly known as taurine, was first isolated in 1827 from the bile of an ox, *Bos taurus*, accounting for its common name (Demarcay, 1838). In phylogenetic terms, taurine is an ancient molecule, because it is found at high concentrations in algae, but absent from most bacteria and viruses, although it has been described as a source of carbon, nitrogen and sulfur in *Bacillus subtilis* (Nakashio et al., 1982). Taurine is found in trace amounts in plants and fungi (Huxtable, 1992). By contrast, it is present at high concentrations in many animals, from insects to mammals, in which it is the most abundant amino acid-related molecule (Huxtable, 1992). Despite being an ancient amino acid, taurine is not incorporated into protein sequences. Moreover, although most taurine is obtained from the diet, this amino acid is considered to be non-essential because it can be synthesized endogenously in the liver of mammals, although this synthesis can be insufficient as in cats (see 1.3.).

1.2. Physicochemical properties

Taurine does not have the classical structure of an amino acid, with an amino moiety in the alpha position of the carboxyl group, but it is nevertheless considered to be an amino acid. Taurine is a free sulfur β -amino acid, in which the amino group is located on the beta-carbon. The molecular structure of taurine ($\text{NH}_3^+-\text{CH}_2-\text{CH}_2-\text{SO}_3^-$) is very similar to that of γ -aminobutyric acid (GABA), the main inhibitory neurotransmitter in the central nervous system, in both the brain and the retina (Fig. 1). Taurine is analogous to β -alanine, a

constituent of vitamin B₅, and to a synthetic compound called guanidoethane sulfonate (GES) (Fig. 1). Both these molecules are competitive inhibitors of taurine to the taurine transporter (Tau-T) activity (see 2.1.2.). Due to the presence of a sulfonic acid, rather than the carboxylic acid group more commonly found in amino acids (Fig. 1), taurine displays specific physicochemical properties, with a very low pK_a value of ~ 2 for its acid group, whereas the pK_a for the amine group is 9, resulting in a zwitterion state at physiological pH (Jacobsen and Smith, 1968; Okaya, 1966). Being taurine highly water soluble, it cannot diffuse across lipophilic membranes. This lack of ability to cross membranes results in steep concentration gradients between intracellular and extracellular compartments (Heinz and Walsh, 1958; Huxtable, 1992; Jacobsen and Smith, 1968), with intracellular concentrations up to 7000 time higher than extracellular concentrations (Piez and Eagle, 1958). Such concentration gradients are generated by an active Na⁺-dependent selective taurine transporter (Tau-T), which takes taurine up into the intracellular compartment. Such uptake processes were initially suspected on the basis of pharmacological studies showing [³H]taurine incorporation into cells (Kishi et al., 1988), and were subsequently definitively demonstrated by the molecular cloning of Tau-T (Liu et al., 1992; Uchida et al., 1993).

1.3. Taurine metabolism

The endogenous synthesis of taurine in various tissues, but mostly the liver and brain (Hayes and Sturman, 1981b; Huxtable, 1989) is one of the sources of this molecule in mammals. The principal metabolic pathway for the synthesis of taurine begins with L-methionine and/or L-cysteine metabolites (Fig. 2). In addition to this principal synthetic pathway, other theoretical biosynthesis routes are possible, making use of enzymatic pathways

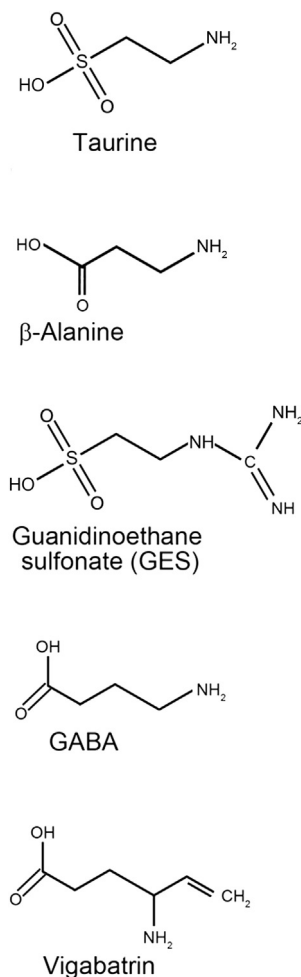


Fig. 1. Chemical structures of taurine and analogous molecules, Chemical formulas of taurine, guanidoethane sulfonate (GES), β-alanine, GABA and vigabatrin (VGB). Note the sulfonic group in taurine and GES, replacing the carboxylic group of β-alanine, GABA and VGB. GES, β-alanine and GABA are competitive substrates or blockers of the taurine transporter (Tau-T). VGB blocks the GABA-transaminase, resulting in an increase in GABA levels and taurine depletion.

linked to the metabolism of sulfur amino acids. In the main pathway, the L-methionine precursor is converted into L-cysteine, via four intermediate steps, involving the formation of S-adenosyl methionine (SAM), S-adenosylhomocysteine (SAH), homocysteine and cystathionine (Fig. 2). Interestingly, cystathionine results from the condensation of one homocysteine molecule with one L-serine molecule, and this reaction is catalyzed by a cystathionine β-synthase. Thus, serine levels can, theoretically, regulate taurine biosynthesis. Finally, cystathionine gives rise to one L-cysteine molecule following the action of two enzymes in succession: cystathionase and cysteine desulfurase (Fig. 2). The L-cysteine generated by this pathway or obtained through the diet can be used as a substrate for taurine generation (Fig. 2). In the major pathway, L-cysteine is converted into cysteine sulfonic acid by cysteine dioxygenase (CDO; Fig. 2). The product of this reaction is then converted into hypotaurine by cysteine sulfonic decarboxylase (CSAD). The concentrations of these two enzymes in tissues are limiting factors, incriminated in the low endogenous synthesis of taurine evidenced in cats see (Hayes and Sturman, 1981b). Three of the enzymes involved in this taurine biosynthesis pathway, cystathionine synthase, γ-cystathionase and CSAD require pyridoxal 5' phosphate (or vitamin B₆) as a cofactor for full enzymatic activity (Fig. 2). Thus a

dietary deficiency of vitamin B₆ can also decrease taurine synthesis, leading to taurine depletion (Yamaguchi et al., 1975).

Taurine anabolism via this major pathway was firstly characterized in the liver (Lombardini et al., 1969), and then in the brain tissues (Tappaz et al., 1992); see (Huxtable, 1992). However, this biosynthetic pathway is considered to be most relevant in the central nervous system (CNS) (Huxtable, 1986). The expression of two key enzymes involved in the synthesis of taurine from cysteine (CDO and CSAD) has been detected in rat brain (Remy et al., 1990). These three enzymes have also been shown to be active in the retina (Macaione et al., 1976). Convincing evidence for taurine synthesis in the CNS was obtained by the use of radioactive [³⁵S]-methionine or -cysteine on rat cortical slices (Tappaz et al., 1992). This synthesis appeared to be restricted to neuronal cell bodies and dendrites, because no CSAD was found in axons (Sturman, 1981). In fact, Dominy et al. (2004) have recently shown that neurones can only produce taurine from hypotaurine whereas astroglial cell can generate the whole synthesis suggesting thereby metabolic interactions between neurones and astroglia for the taurine supply. Hepatic and cerebral tissues differ in their ability to synthesize taurine, and this ability is also affected by age and species. Indeed, for all species studied to date, a lower capacity for taurine synthesis has been found in the liver and brain of young animals than in those of adult animals (Loriette and Chatagner, 1978). This observation reflects the limited CSAD activity and accounts for the need to supplement baby food with taurine (Gaull, 1989). For adults of species such as cats, monkeys and humans, CSAD activity levels are extremely low in the liver, limiting the endogenous synthesis of taurine and rendering the animals dependent on dietary taurine intake. By contrast, rats and dogs display higher levels of activity for this enzyme and are therefore less dependent on dietary taurine intake. For example, 80% of the cysteine pool is converted to taurine in rats, versus only 20% in cats. Hence, taurine biosynthesis cannot account for the physiological taurine concentrations in tissues in any of the species studied. Dietary taurine intake is therefore essential for taurine homeostasis.

The other two minor routes of taurine synthesis involve (i) cysteic acid, which is synthesized from cysteine sulfonic acid to generate taurine directly (Fig. 2), and (ii) cysteamine, which is produced from cysteine to give hypotaurine (Fig. 2). The first minor route (the “cysteic acid route”) is functional in the brain and liver (Huxtable, 1989) whereas the second minor route (the “cysteamine route”) is active principally in the mammalian kidney (Dupre and De Marco, 1964) and inactive in brain, in which cysteamine is present in only very small amounts (Huxtable, 1989).

Approximately 25% of free taurine is eliminated directly in the urine (Hayes and Sturman, 1981b). Taurine catabolism is dependent principally on conjugation with cholic acid to form bile acids in the liver (Hansen, 2004), but taurine can also form Schiff bases with aldehydes and ketones (Hansen, 2004). Moreover, the formation of N-chlorotaurine (or taurine chloramine), mostly in neutrophils, results from the association of taurine with hypochlorous acid (HOCl), catalyzed by a myeloperoxidase (see Schaffer et al., 2009).

1.4. Mammalian taurine contents

1.4.1. Body content

Intracellular free taurine content is high in all mammals and this compound is supplied to all tissues (Huxtable, 1992; Ripps and Shen, 2012). However, the taurine contents of the plasma and tissues are highly species-dependent. Overall, taurine levels appear to be lower in primates than in rodents or rabbits (Table 1), probably due to the lower capacity for taurine synthesis in primates. Thus, plasma taurine concentrations may range from 80 μM to 770 μM, as a function of species (Table 1). In a recent study on a cohort of 69

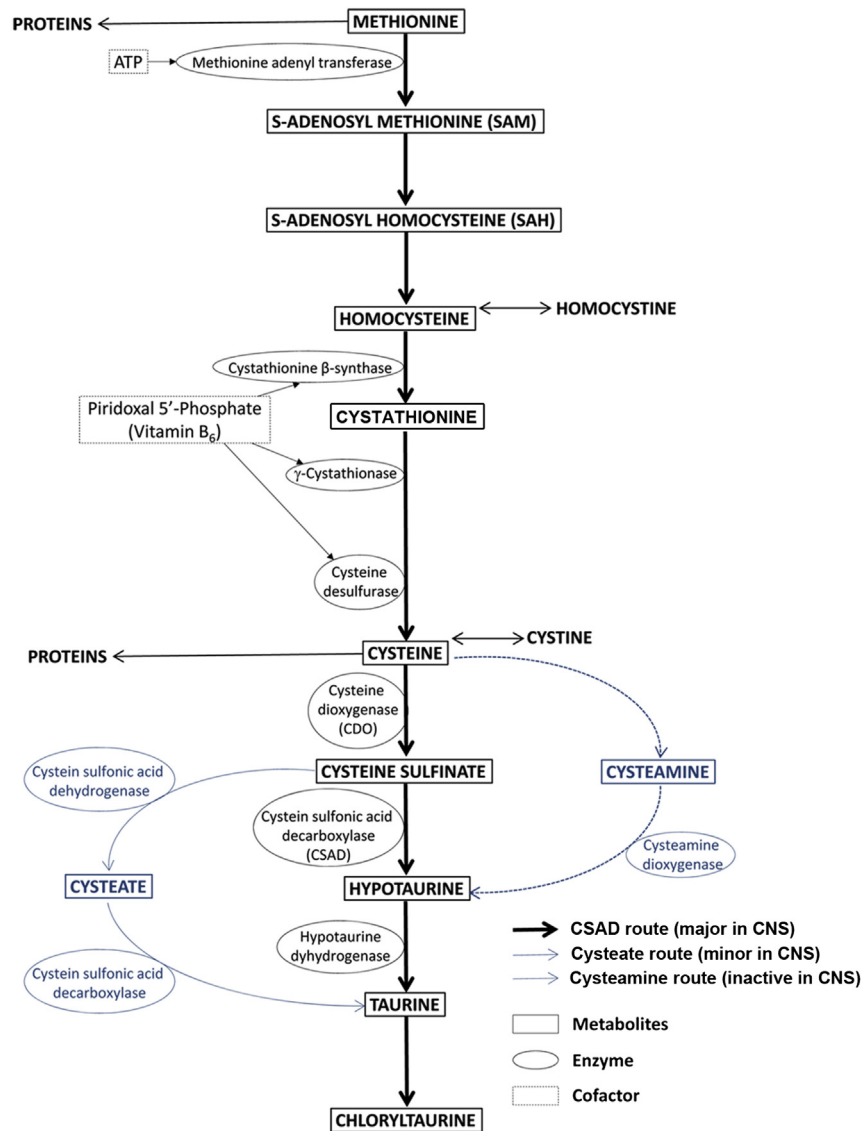


Fig. 2. The metabolic pathway for sulfur amino acids leading to taurine biosynthesis in mammalian liver, kidney and brain. The “CSAD route”, represented as a bold black arrow, is the major route of taurine biosynthesis in the CNS, liver and kidney. The “cysteate route”, represented as thin blue arrows, constitutes a minor route of taurine biosynthesis in the CNS while the “cysteamine route”, represented as a blue hatched arrow, is not active in the CNS.

pet dogs, the mean of plasma taurine concentration while 162 μM (Rosolen et al., 2013). In adult tissues, taurine concentrations are high in skeletal muscles, which contain the largest pool of taurine (Huxtable, 1992). Furthermore, all vital organs, including the central nervous system (CNS), heart, liver and kidney, also contain high concentrations of taurine (Hayes and Sturman, 1981b), in the range of 5–30 mmol/g tissue (Hansen, 2001).

1.4.2. Taurine content of the eye

Taurine is present in large amounts in the eyes, but its physiological role remains unclear. Indeed, it is one of the most abundant amino acids in ocular tissues, such as the cornea, iris, lens and ciliary bodies (Ripps and Shen, 2012). Taurine concentrations of about 10 $\mu\text{mol/g}$ dry tissue have been obtained for the ciliary bodies (Heinamaki et al., 1986; Reddy, 1967, 1968), and taurine has also been found at very high concentrations in ocular fluids, such as the rat vitreous humor (Heinamaki et al., 1986). We recently showed that mean taurine concentration in the aqueous humor of pet dogs can reach up to 50 μM (Rosolen et al., 2013). The amino-acid

precursors of taurine biosynthesis (i.e. methionine and cysteine; see Fig. 2) are also present in these eye structures. However, methionine is present at concentrations between 0.09 and 0.37 $\mu\text{mol/g}$ dry tissue in the eye, whereas cysteine is present only in trace amounts, except in the retina (Heinamaki et al., 1986). Few

Table 1

Taurine concentrations in the plasma and tissues of rodents and primates.

Species	Plasma (μM)	Tissues ($\mu\text{mol/g}$ wet wt)			
		Retina	Brain	Heart	Liver
Rat	222–450	50	3–9	27	3
Mouse	740–770	16–40	9	43	–
Monkey	100–190	28–31	2–4	11	3
Human	80	–	2	6	2

Taurine concentrations were measured in adults, except for the monkey retina values, which were obtained from five-month-old animals. Values were reported in Hayes and Sturman (1981b), except for the murine plasma taurine concentrations, which were reported in our studies Froger et al. (2012), Gaucher et al. (2012) and Jammoul et al. (2009).

data describing the functional role of taurine in the various eye structures are available. Taurine uptake in the lens is weak, suggesting potential endogenous biosynthesis rather than exogenous uptake from dietary sources. A potential role in cataract formation has been suggested, because low taurine contents have been reported in human patients with age-related cataracts and in rats with galactose-induced cataracts (Gupta and Mathur, 1983). *In vitro*, incubation with taurine has been shown to limit damage in a model of diabetic cataracts (Kilic et al., 1999). Taurine uptake into the cornea is also weak, suggesting that endogenous production may be involved in generating the levels of 10 $\mu\text{mol/g}$ dry tissue observed. In addition, taurine plays a role in the osmoregulation of the corneal epithelium contributing thereby to the prevention of hypertonicity-induced dry eye syndrome (Shioda et al., 2002). The vitreous humor probably serves as a medium for ionic exchanges between the anterior segment and the retina and for lymphatic drainage. As an organic osmolyte, taurine may regulate these exchanges, but its exact role in the vitreous humor remains unknown.

In the mammalian retina, taurine is the most abundant amino acid before glutamate, during both development and adulthood (Macaione et al., 1974). Moreover, the retina appears to be the taurine richest organ (Ripps and Shen, 2012), with concentrations greater than those in other ocular structures or in the brain, in adult animals of all species examined (Pasantes-Morales, 1985; Pasantes-Morales and Cruz, 1985b), reaching up to 50 $\mu\text{mol/g}$ tissue in rats (Table 1; see Huxtable, 1989). Even though the retinal concentrations of taurine have long been known to be very high and many studies have investigated the function of this amino acid, the specific role of taurine in the retina remains unclear.

1.4.3. Taurine content during development

During mammalian development, retinal taurine levels seem to be highest in neonates, gradually falling to one third the levels of birth during adulthood (Sturman, 1988). Quantitative measurements have indicated that pups receive a significant proportion of this taurine from maternal milk (Sturman, 1988). Macaione et al. (1974) reported that taurine levels in rat neonates gradually increase to reach a peak at 50 $\mu\text{mol/mg}$ proteins at the age of 30 days, subsequently decreasing and stabilizing at 40 $\mu\text{mol/mg}$ proteins, by the age of 45 days. Even higher concentrations were reached in mice, exceeding 100 $\mu\text{mol/g}$ tissue (Sturman, 1988). This early increase in taurine concentration coincides with the formation of the photoreceptor layer. This layer accounts for more than 60% of all the taurine present in the retina (Huxtable, 1989), in which taurine is specifically located in the outer nuclear layer (Pasantes-Morales et al., 1972). The presence of such high concentrations of taurine amounts in photoreceptors was confirmed in animal models of retinitis pigmentosa (RP), with photoreceptor degeneration associated with a decrease in taurine levels to 25% the peak value (Schmidt, 1981). During aging, plasma taurine concentration decreases and tissue contents have also been shown to decline in all organs examined (liver, spleen, kidney, heart, skeletal muscles, brain and eyes) (Dawson et al., 1999), suggesting that aging may be associated with the development of taurine deficiency.

1.5. Taurine and nutrition

As pointed out above, endogenous taurine synthesis alone cannot account for the physiological concentrations of taurine observed in tissues and exogenous taurine must therefore be obtained from food. As taurine is almost exclusively limited to the animal kingdom, meats (particularly uncooked meat), seafood and fish are the major sources of taurine (Table 2; Zhao, 1994). For example, taurine content may reach up to 30 $\mu\text{mol/g}$ (wet wt) in beef and 9 $\mu\text{mol/g}$ (wet wt) in lamb. Some of the taurine

Table 2
Taurine content of various foods.

Seafood and fish	Taurine content (mg/100 g)	Meat	Taurine content (mg/100 g)
Octopus	390	Beef muscle	50–100
Prawn	143	Beef liver	42
Shrimp	115	Uncooked lamb muscle	310
Tuna	332	Cooked Lamb muscle	171
Sole	256	Pork muscle	118
Carp	90	Pork liver	42
Crab	278	Chicken	378

Values are expressed in mg of taurine per 100 g food. Data from Purchas et al. (2004) and Zhao (1994).

concentrations reported for fish are remarkably high: up to 83 $\mu\text{mol/g}$ (wet wt) in yellowtail (Japanese amberjack) and 78 $\mu\text{mol/g}$ (wet wt) in mackerel. Milk and eggs also have high taurine contents. Indeed, taurine concentrations in milk have been estimated at $\sim 600 \mu\text{M}$ in gerbil, $\sim 300 \mu\text{M}$ in cat and $\sim 40 \mu\text{M}$ in humans (Hayes and Sturman, 1981a). By contrast, taurine is absent, or present only in trace amounts in vegetables and mushrooms (Huxtable, 1992). Cats and primates (including humans) are known to have a limited capacity for taurine synthesis, and the feeding of infant monkeys with plant-derived milk substitutes, such as soy milk, can result in significant taurine depletion (Sturman et al., 1988).

Taurine intake from dietary sources is highly dependent on taurine transporter expression in tissues. Dietary taurine must first be transported across the intestinal barrier to reach the blood. Humans have two intestinal transport mechanisms that mediate the transfer of taurine across the brush-border membrane of the intestine (Anderson et al., 2009). The first one involves a high-affinity, low-capacity Na^+ - and Cl^- -dependent taurine transporter (Tau-T; SLC6A6 gene). Tau-T has been isolated and cloned from a number of tissues and from various species, including rat, mouse and human (Liu et al., 1992; Smith et al., 1992; Uchida et al., 1993). The second mechanism, as suggested by early studies, involves a low-affinity transport for amino-acids and represents the major uptake for taurine (Munck and Munck, 1992). An H^+ -coupled transporter recognizing a broad range of amino acids has been identified at the brush-border membrane of human intestinal cells. This transporter, the proton-coupled amino-acid transporter (PAT1) (Thwaites et al., 1995) has been found in rat, mouse and human tissues, but is absent from the small intestine of rabbit and guinea pig, in which the high-affinity Tau-T remains the only known mechanism for transporting taurine of dietary origin (Munck and Munck, 1992, 1994). Anderson et al. (2009) recently assessed the differences in function between PAT1 and Tau-T. They suggested that PAT1 might be responsible for bulk taurine uptake during meals, with Tau-T instead involved in active taurine uptake into the intestinal epithelium between meals, even at low concentrations.

1.6. Taurine and cellular physiology

1.6.1. Regulation of osmolarity

The osmoregulatory properties of taurine underlie its principal physiological action in the control of cell volume. In the brain, taurine may account for 50% of the osmolytes required for the regulation of cell volume (Trachtman et al., 1988). Other studies have shown that taurine is the main molecule supporting cellular efflux in astrocytes (Beetsch and Olson, 1998). Accordingly, two different mechanisms are involved in taurine transport: external taurine concentration and external osmolarity (Jones et al., 1995). Finally, hyperosmolarity increases the production and activity of Tau-T in retinal cells (El-Sherbeny et al., 2004).

1.6.2. Antioxidant properties

It is now widely accepted that oxidative stress plays a crucial role in the development of many chronic diseases, including cardiovascular diseases, such as arteriosclerosis (Magenta et al., 2013), metabolic syndromes, such as diabetes mellitus (Stadler, 2012) and brain diseases, such as Alzheimer's disease (Yan et al., 2013). In the retina, oxidative stress has also been incriminated in the development of many degenerative processes, in age-related macular degeneration (Beatty et al., 2000) and glaucoma (Chrysostomou et al., 2013), for example. Oxidative stress leads to the overproduction of highly reactive free radicals, known as reactive oxygen species (ROS), the classical products of which are the superoxide anion (O_2^-), the hydroxyl radical (OH), and hydrogen peroxide (H_2O_2). Taurine is considered to be an antioxidant, but the cellular mechanisms underlying its antioxidant properties have never been clearly characterized, particularly in retinal cells.

The sulfonic group may confer antioxidant properties by directly neutralizing ROS production. However, such direct scavenging actions against classic ROS production are generally attributed to hypotaurine rather than taurine (Aruoma et al., 1988). However, taurine was shown to neutralize hypochlorous acid (Schaffer et al., 2009; Timbrell et al., 1995), nitric oxide (Redmond et al., 1996), and even hydrogen peroxide, directly, in one *in vitro* study (Cozzi et al., 1995). Otherwise, the antioxidant action of taurine involves indirect mechanisms leading to an attenuation of the deleterious effects of ROS. Its osmotic properties could also contribute to limit the effect of ROS-induced plasma membrane permeabilization. These antioxidant action also result from the control of ion and water efflux by taurine after membrane alterations (Timbrell et al., 1995; Wright et al., 1985). Hamaguchi et al. (1991) have suggested that the interaction of taurine with the plasma membrane results in membrane lipid content (phosphatidylethanolamine versus phosphatidylcholine), leading to a decrease in membrane fluidity and an increase in resistance to oxidative challenge. However, one major antioxidant effect of taurine has already been demonstrated: the capacity of this molecule to maintain antioxidant enzyme levels after exposure to toxins. The enzymes up-regulated by taurine include thioredoxin reductase (Yildirim et al., 2007), glutathione peroxidase (Flora et al., 2004) and superoxide dismutase in the brain (Nonaka et al., 2001; Vohra and Hui, 2001). Finally, the precise mechanisms underlying the antioxidant properties of taurine remain hypothetical and further investigations are required, particularly in retinal tissue, in which oxidative stress plays a major role in the development of disease.

1.6.3. Calcium modulation

Another important function of taurine is the regulation of cellular calcium (Ca^{2+}) concentrations (Foos and Wu, 2002; Schaffer and Azuma, 1992), which may also be cytoprotective. Indeed, taurine modulates numerous Ca^{2+} -dependent processes in the heart, brain and retina (Huxtable, 1989). This regulation is linked to the capacity of taurine (i) to increase high-affinity Ca^{2+} binding without affecting low-affinity binding and (ii) to inhibit Ca^{2+} uptake significantly (Lazarewicz et al., 1985). Taurine can also prevent calcium influx into neurons through its direct action on L-, P/Q- and L-type calcium channels (Wu et al., 2005). It has also recently been shown in chromaffin cells that taurine regulates voltage-dependent calcium channels by stimulating metabotropic-like glycinergic receptors (Albinana et al., 2010). All these actions converge on the inhibition of Ca^{2+} influx into the cytosolic compartment.

The modulation of calcium influx by taurine may account for the neuroprotective role of this compound in preventing excitotoxicity due to excess glutamate release and glutamate receptor activation (El Idrissi and Trenkner, 1999). We have recently shown that taurine

can prevent glutamate excitotoxicity in NMDA-exposed retinal ganglion cells (RGCs) (Froger et al., 2012). The role of taurine in limiting intracellular calcium accumulation (El Idrissi and Trenkner, 1999) may also contribute to its neuroprotective effects because during glutamate excitotoxicity, Ca^{2+} accumulation in mitochondria was shown to trigger to an excessive ROS generation (Vesce et al., 2004).

1.6.4. Taurine-elicited neurotransmission

In the retina and, more generally, in the CNS, taurine is considered to be an agonist of all GABA receptors, including the GABA_{A/C} ionotropic and GABA_B metabotropic receptors (Albrecht and Schousboe, 2005; Jones and Palmer, 2009). However, the taurine-elicited of GABA_B receptor activation remain somehow controversial because no clear evidence to demonstrate that taurine is a GABA_B receptor agonist. It has also been suggested that taurine activates the strychnine-sensitive glycine ionotropic receptors in RGCs (Bulley and Shen, 2010) and in cone photoreceptors (Balse et al., 2006). One recent study also showed taurine-induced 5HT receptor activation in retinal neurons (Bulley et al., 2013). The existence of specific taurine receptors has been suggested by several authors, but such receptors have never been clearly identified.

1.7. Taurine in non-retinal diseases

It has been suggested that dietary taurine supplementation could be used for treatment of many diseases. Indeed, this amino acid has been shown to regulate blood pressure and has therefore been proposed as an antihypertensive agent (Militante and Lombardini, 2002). It has also been associated with an improvement of other cardiovascular conditions, including stroke and cardiac hypertrophy (Dawson et al., 2000; Yamori et al., 1996). Taurine has also been evaluated as a treatment for alcoholism, because it has been shown to decrease the frequency of psychotic events during withdrawal (Ikeda, 1977) and to reduce the hepatic damages induced by ethanol consumption in animals (Wu et al., 2009). Indeed, many therapeutic benefits of taurine supplementation have been described on myotonic dystrophy (10 g/day for 6 months), congestive heart failure (6 g/day for 4 weeks), borderline hypertension (6 g/day for 7 days), type II diabetes (3 g/day for 4 months), overweight, cystic fibrosis (500–1500 mg/day for 1 year) and anemia (1 g/day for 20 weeks) (Shao and Hathcock, 2008). Furthermore, no adverse effects or relevant biochemical changes have been observed following taurine supplementation (1.5 g/day for 8 weeks) (Brons et al., 2004; Spohr et al., 2005).

2. Taurine in retinal physiology and diseases

2.1. Taurine and photoreceptors

2.1.1. Photoreceptor degeneration on a taurine-free diet

A potential role of taurine in retinal physiology was first suggested following the observation of retinal damage in cats fed a taurine-free diet (Hayes et al., 1975). As mentioned above, endogenous taurine biosynthesis is not sufficient to meet the physiological needs of the retina *in situ*. Indeed, the documented lower activity of CSAD, a key enzyme in taurine biosynthesis (Fig. 2), in the liver of cats and primates (including humans) may reflect lower levels of biosynthesis activity in the retinas of these animals (Lombardini, 1991), suggesting a reliance on dietary taurine to maintain physiological levels of this compound in the retina. The distribution of Tau-T *in situ* in the retina suggests that taurine of dietary origin present in the plasma is taken up (i) by the retinal pigment epithelium (RPE) in the outer retina to supply

the photoreceptors, or (ii) by the capillary endothelium in the inner retina for transfer to the ganglion cell layer (Vinnakota et al., 1997).

Accordingly, cats fed exclusively on casein, as a taurine-free diet, display massive photoreceptor loss (Hayes et al., 1975). Further studies showed similar photoreceptor degeneration in cats with prolonged nutritional taurine deprivation (Aguirre, 1978; Berson et al., 1976; Schmidt et al., 1976). Amino acid analyses revealed that this degeneration was strongly associated with a decrease in plasma and retinal taurine concentrations (Hayes et al., 1975). The retinal lesions occurring in cats fed a taurine-free diet were specifically assessed by functional investigations and histological analysis. Electroretinogram (ERG) recordings showed smaller amplitudes of both cone and rod responses. In addition, a delayed implicit times for cones were evidenced suggesting a slower cone response to the light in these cats (Berson et al., 1976; Schmidt et al., 1976). These findings revealed that taurine-free diet in cats induced an alteration of the retinal function, which was characterized by lower and slower visual ERG signals.

The histological consequences of taurine deprivation have been well documented in studies showing central and peripheral retinal damage, resulting in retinal lesions with an omega shape, which can be observed either *in vivo* or *in situ* (Leon et al., 1995). These lesions range in size from small areas of focal atrophy located in the central area of the retina (feline central retinal degeneration) to generalized retinal atrophy (Aguirre, 1978; Leon et al., 1995). The outer layers were the most strongly affected, with disorganization of the outer segments of the photoreceptors leading to cell death. This suggests that the photoreceptors are the retinal neurons most sensitive to taurine deficiency. At later stages, histological examination revealed a surprising displacement of photoreceptor cell bodies into the subretinal space. Such tissue damage was observed when plasma taurine concentrations fell by a factor of at least two (Schmidt et al., 1977).

Similar findings were obtained in experiments on baby primates (*Cebus* and cynomolgus monkeys) raised from birth on soybean infant milk formula lacking taurine. This diet greatly decreased plasma taurine concentrations (Hayes et al., 1980), contributing to a loss of visual acuity associated with morphological changes (Imaki et al., 1987). The structural modifications observed included degenerative ultrastructural changes in the photoreceptor outer segments ranging from swelling and disorientation to fragmentation and disorganization, with cones more severely affected than rods because the changes were most pronounced in the foveal region (Imaki et al., 1987).

In humans, both children and adults subjected to long-term parenteral nutrition lacking taurine have been shown to display lower fasting plasma taurine concentrations than controls. Consequently, most of the patients examined were shown to have abnormal ERG patterns (Ament et al., 1986; Geggel et al., 1985), which were attributed to the taurine deficiency because an intravenous solution of taurine restored normal plasma taurine levels and ERG findings in some patients (Ament et al., 1986; Geggel et al., 1985). Taurine deficiency has also been reported in vegans (i.e. individuals eating no animal products) (Laidlaw et al., 1988; Rana and Sanders, 1986) and visual deficits have also been reported in this context (Milea et al., 2000). These findings thus establish the importance of dietary taurine for the maintenance of normal retinal function.

2.1.2. Taurine depletion induced by pharmacological treatments

Taurine deficiency may also be achieved by the prolonged treatment of animals with drugs selectively blocking Tau-T activity. The principal pharmacological agents used for this purpose are β -alanine and guanidoethane sulfonate (GES). β -alanine is a natural

compound included in the composition of peptides such as carnosine, which is known to reduce muscle fatigue in athletes. It also forms part of several other important molecules, such as pantothenic acid (vitamin B₅), but is not present in proteins. Both GES and β -alanine have been shown to decrease (i) the assimilation of exogenous taurine of dietary origin and (ii) taurine uptake into tissues and cells. Measurements of [³H]taurine uptake in isolated heart tissue and retinal tissues have shown that β -amino acids can inhibit taurine uptake in a specific and competitive manner, whereas α -amino acids display no affinity for Tau-T. L-glutamate and L-cysteine have activity at millimolar concentrations, but L- β -alanine inhibits Tau-T in the retina with a much higher efficiency (Starr and Voaden, 1972). GES, a structural analog of taurine (Fig. 1) discovered by the team of Prof. Ryan J Huxtable, is also a powerful inhibitor of Tau-T (Huxtable and Lippincott, 1981). The administration of GES for four weeks has been shown to decreased taurine levels by 80% in the heart, 76% in the liver and 67% in the cerebellum (Huxtable et al., 1979). In the retina, taurine depletion induced by chronic GES treatment or by β -alanine treatment in rats has been shown to induce morphological alterations in photoreceptors, the changes observed being most marked in the outer segments (Lake and Malik, 1987; Pasantes-Morales et al., 1983), as initially reported in cats fed a taurine-free diet (Hayes et al., 1975). We recently confirmed these results with GES, by showing further that cone photoreceptors were much more sensitive to taurine depletion than rods (Gaucher et al., 2012).

2.1.3. Genetic deletion of the taurine transporter, Tau-T

Taurine is taken up into cells by a Na⁺-dependent active transporter, the taurine transporter, Tau-T. Taurine can then carry out multiple functions in cells, in osmoregulation, antioxidant defense, protein stabilization, stress responses, neuromodulation and immunomodulation, for example, thereby contributing to the protection of the cell against various types of injury (Huxtable, 1992). For further characterization of the contribution of taurine to various cellular functions and of its role in disease, a mutant knockout mouse, in which the gene encoding the taurine transporter was disrupted (Tau-T KO mice), was generated, resulting in an animal model with permanent taurine deficiency (Heller-Stilb et al., 2002; Ito et al., 2008). As expected, these mice showed very low taurine concentrations in various organs, including the eyes, in which taurine levels were about 74% lower than those in control wild-type mice (Heller-Stilb et al., 2002). The most prominent morphological feature of Tau-T KO mice is severe, progressive retinal degeneration, which begins as soon as two weeks after birth, when the eyes first open (Heller-Stilb et al., 2002; Rascher et al., 2004). Histological experiments revealed a deterioration of the inner and outer photoreceptor segments and a gradual decrease in ONL thickness, with the induction of apoptosis in the ONL, as indicated by TUNEL assays. At the age of one to two months, photoreceptor cell degeneration was highly advanced, with most of the remaining nuclei pyknotic with detectable ribbon synapses and the presence of occasional remnants of outer segment disks. After two months, the photoreceptors had disappeared, such that the neurons of the inner nuclear layer (INL) lined the RPE. At advanced stages of the degenerative process, the INL was irregular, varying in thickness from only a few cells to several cell layers with a clear inner plexiform layer and a ganglion cell layer (Heller-Stilb et al., 2002). ERG measurements demonstrated cellular activity in two-week old animals, but with wave amplitudes 70% smaller than those in control wild-type mice, with a further reduction to 10–15% control values at three weeks (Heller-Stilb et al., 2002). After the age of six weeks, the ERG was completely flat in Tau-T KO mice, confirming the loss of both rods and cones (Heller-Stilb et al., 2002). These results are consistent with taurine and its transporter playing

a major role in the maintenance of normal retinal function and morphology.

2.1.4. Phototoxicity and taurine deficiency

Mammalian photoreceptors are stimulated by visible light at wavelengths of 400–760 nm. However, light has been identified as a risk factor in various eye diseases and during exposure to intense light (Behar-Cohen et al., 2011). In particular, shorter wavelengths interacting with chromophores located in photoreceptors and retinal pigment epithelial (RPE) cells can cause oxidative stress and severely damage these cells (Hunter et al., 2012; Rozanowska, 2012; van der Torren et al., 2002). Among these chromophores, A2E (N-retinylidene-N-retinylethanolamine) was found to accumulate in the lipofuscin formed in RPE cells during the degradation of photoreceptor outer segments. Recently, wavelengths of 415–455 nm was recently defined as the most toxic forms of visible light in the solar spectrum reaching the retina for A2E-loaded RPE cells (Arnault et al., 2013).

In the 1980 and 1990s, several investigations on taurine-deprived animals demonstrated an interaction between light exposure and observed photoreceptor degeneration. This increase in phototoxicity in taurine-deprived animals was first described in animals on long-term GES treatment, in studies examining retinal function and morphology after exposure to various types of lighting conditions (Cocker and Lake, 1989; Quesada et al., 1988; Rapp et al., 1988). The decrease in amplitude of both the ERG a- and b-waves was more pronounced in rats exposed to light than in those kept in the dark (Quesada et al., 1988). Similar results were published by Rapp et al. (1988) for albino pups raised from birth in either dim (2 lux) or brighter (300 lux) light conditions (with light/dark cycles) and deficient in taurine due to the supply of drinking water containing 1% GES from weaning. Dim light induced decreases of the ERG a- and b-wave amplitudes of 36 and 46%, respectively, whereas the highest light intensity decreased these amplitudes by 94 and 89%, respectively. The rats exposed to the highest light intensity displayed 62% photoreceptor loss, with the greatest cell losses occurring in the upper central area of the retina (Rapp et al., 1988). Similarly, greater photoreceptor degeneration was observed in Taut-T KO mice exposed to light than in those kept in the dark (Rascher et al., 2004), although animals kept in total darkness nevertheless displayed some photoreceptor damage (Rascher et al., 2004). These studies demonstrated that deficiency in taurine increases susceptibility to light.

The morphological disruption of the retina observed after exposure to high-intensity light was characterized by the distension of outer segment discs and a marked swelling of rod outer segments not seen in dim illumination conditions and effectively counteracted by taurine supplementation (Pasantes-Morales et al., 1981; Pasantes-Morales and Cruz, 1985a). Light exposure increases retinal lipid peroxidation and reduces both superoxide dismutase and glutathione peroxidase activities, so the protective effect of taurine against phototoxicity has been attributed mostly to the antioxidant properties of this molecule (Pasantes-Morales and Cruz, 1985a; Yu et al., 2008). Exposure to light seems to modulate taurine cellular homeostasis by decreasing the intracellular uptake of taurine and stimulating taurine release (Pasantes-Morales et al., 1973a, 1973b). Furthermore, light may also modulate taurine synthesis, because a decrease in retinal CSAD activity has been reported in frogs exposed to light (Ida et al., 1981). Exposure to light has been shown to modify retinal taurine synthesis (Nishimura et al., 1983). This effect may account for the lower retinal taurine content described in light-exposed rats (Oraedu et al., 1980). The synergism between environmental light and taurine depletion in photoreceptor degeneration may therefore result from a light-induced decrease in taurine content, leading to lower levels of

antioxidant protection and, in parallel, an increase in photochemical stress.

2.1.5. Taurine in retinal diseases with photoreceptor loss

The demonstration of a major role for taurine in photoreceptor function has suggested that this molecule may be involved in diseases with photoreceptor degeneration. Various studies have investigated whether taurine levels contribute to photoreceptor degeneration in retinitis pigmentosa (RP), a heritable retinal dystrophy. One study showed that the onset of photoreceptor cell degeneration preceded taurine loss in the retina in a dog (Irish Setter) model of rod-cone dysplasia (Schmidt and Aguirre, 1985). These observations are consistent with previous findings in two animal models of RP — mice with retinal dystrophy (Orr et al., 1976) and Royal College of Surgeons rats (Schmidt and Berson, 1978) — confirming that retinal taurine levels fall after the loss of photoreceptor cells (Matuk, 1984).

In addition, RP patients have lower levels of taurine uptake into platelets (Airaksinen et al., 1981, 1979) and lower blood taurine concentrations (Airaksinen et al., 1981; Arshinoff et al., 1981) than healthy patients, but taurine concentration is not considered to be a reliable biomarker for RP (Benini et al., 1988). Nevertheless, taurine supplementation has been proposed as a treatment for the preservation of photoreceptors in RP (Reccia et al., 1980). A double-blind controlled study testing a combination of taurine supplementation and treatment with diltiazem and vitamin E concluded that such treatment slowed the progression of vision deficits in RP patients (Pasantes-Morales et al., 2002).

2.2. Taurine and retinal ganglion cells (RGCs)

2.2.1. Early studies

Various studies have assessed the consequences of taurine depletion due to either dietary deprivation (Hayes et al., 1975) or pharmacological treatments with a Tau-T blocker (Pasantes-Morales et al., 1983), but only one of these studies investigated the fate of RGCs (Lake et al., 1988). In this study, RGCs and their axons gradually disappeared in GES-treated animals, but this RGC loss was considered secondary to photoreceptor degeneration, as in animal models of RP and human patients with this disease. Imaki et al. (1998) in a study on cats treated by β -alanine for 40 weeks also concluded that cells in the inner retina, including RGCs, were not substantially affected by the taurine depletion, although they noted “varying degrees of the decline of both the number and size of ganglion cells and blood vessels”. Similarly, investigations of retinal tissue from Tau-T KO mice did not clearly assess the integrity of the GCL (Heller-Stilb et al., 2002; Rascher et al., 2004). However, the expression of Tau-T by RGCs was suggested by a study on an immortalized RGC cell line (El-Sherbeny et al., 2004), and shown in a recent study of purified RGCs from adult rats (Froger et al., 2012; see Fig. 6). Taurine was also reported to protect this cell line from hypoxia-induced apoptosis (Chen et al., 2009). We recently re-examined RGC loss in taurine-depleted animals, after our studies on vigabatrin-treated animals (see below). We found that RGC cell loss occurred in parallel to cone photoreceptor loss in GES-treated mice (Gaucher et al., 2012).

2.2.2. The retinal toxicity of vigabatrin

γ -vinyl- γ -amino butyric acid or Vigabatrin (VGB) is an anti-epileptic agent with a chemical formula very similar to those of GABA and taurine (Fig. 1). Our study on the retinal toxicity of VGB led us to investigate the role of taurine in RGC survival. VGB is used to treat epileptic seizures by irreversibly inhibiting GABA transaminase, the GABA-degrading enzyme found in astrocytes, thereby increasing the brain concentrations of GABA (Grant and Heel, 1991),

the main inhibitory neurotransmitter of the CNS. The consecutive increase in intracellular GABA hampers GABA uptake resulting thereby in an increase in extracellular GABA. Chronic VGB administration has been shown to induce an irreversible bilateral constriction of the field of vision in 10–40% of treated patients (Eke et al., 1997; Krauss et al., 1998; Ruether et al., 1998). This visual field loss is associated with more central alterations of visual acuity (Hillenkamp et al., 2004; Johnson et al., 2000), color discrimination (Hilton et al., 2002) or contrast sensitivity (Hillenkamp et al., 2004; Miller et al., 1999). The imaging of retinas from patients has demonstrated the occurrence of optic atrophy, with a loss of ganglion cell fibers, following observations of retinal sections by optical coherence tomography (OCT) or of eye fundi (Buncic et al., 2004; Frisen and Malmgren, 2003; Wild et al., 2006). The lower amplitudes of photopic ERG (see Fig. 3) and flicker responses were also indicative of cone photoreceptor damage (Buncic et al., 2004; Miller et al., 1999; Westall et al., 2003). The smaller amplitudes observed on electro-oculograms were indicative of an underlying dysfunction of RPE/photoreceptor interactions (Arndt et al., 1999; Lawden et al., 1999). Other studies have also reported decreases in the amplitude of the scotopic ERG b-wave, suggesting more complex impairment of the retinal network (Daneshvar et al., 1999). A post-mortem histological study on patients showed that the GCL was the primary site of lesions (Ravindran et al., 2001). Despite this retinal toxicity, VGB remains the first-line treatment for infantile spasms and it was even recently accepted for this indication by the American FDA (Ben-Menachem et al., 2008; Chiron et al., 1997). It is also prescribed as a third-line treatment for drug-resistant epilepsy.

In a recent study on VGB-treated rats, we found that retinal lesions were associated with taurine depletion, with plasma

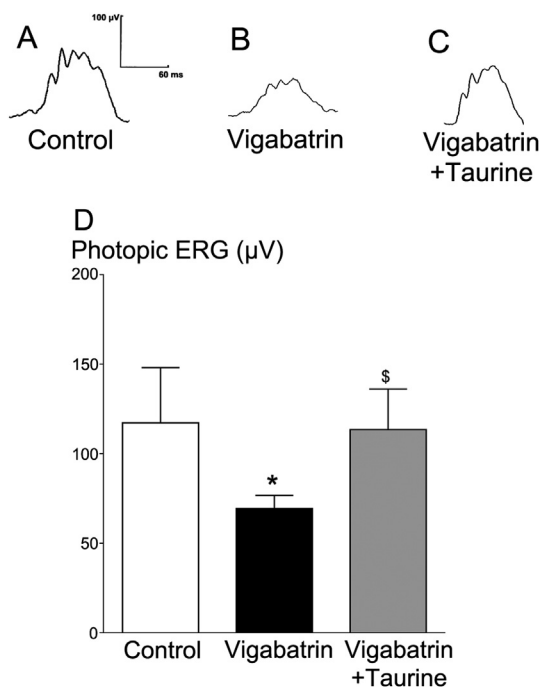


Fig. 3. Functional rescue by taurine supplementation in VGB-treated rats. A–C: Representative photopic ERG from left eye of rats treated during 45 days with either vigabatrin (50 mg/kg, i.p.; B) or vigabatrin (50 mg/kg, i.p.) plus taurine (420 mg/kg, i.p., C), as compared to control rats treated with saline (0.09%, i.p.; A). D: Quantification of photopic ERG amplitudes measured from rat treated with either vigabatrin (black bar, $n = 5$) or vigabatrin plus taurine (gray bar, $n = 5$), as compared to control rats treated with saline (white bar, $n = 10$). The data are means $\pm$ SEM from independent rats. * $p < 0.05$ as compared to control group and \$ $p < 0.05$ as compared to vigabatrin-treated group (One-way ANOVA followed by Bonferroni post hoc test).

taurine levels being 67% lower than those in control rats (Jammoul et al., 2009). This finding was extended to patients treated for infantile spasms. For example, one patient with a normal plasma taurine concentration before treatment had taurine levels below the detection threshold after 15 months of treatment (Jammoul et al., 2009). Furthermore, all the retinal damages observed in VGB-treated rats and mice were abolished by taurine supplementation in drinking water (Jammoul et al., 2010, 2009). Retinal damages were observed in albino rats or mice after 25–60 days of VGB treatment. ERG recordings have shown that VGB treatment in newborn rats (P4) treated for 45 days induced a significant alteration of photopic ERG (Fig. 3B, D) as compared to untreated rats (Fig. 3A, C). Interestingly, taurine administration together with VGB allowed prevention of this deficit in retinal function (Fig. 3C, D). Histological investigations displayed that retinal injuries included RGC loss (Jammoul et al., 2010), cone photoreceptor damage as indicated by the reduced number of their outer segments (Duboc et al., 2004; Jammoul et al., 2010, 2009), retinal gliosis (Duboc et al., 2004; Jammoul et al., 2010, 2009; Ponjavic et al., 2004), rod bipolar cell sprouting (Jammoul et al., 2010, 2009; Wang et al., 2008) and disorganization of the outer nuclear layer (ONL) (Butler et al., 1987; Duboc et al., 2004; Jammoul et al., 2010, 2009; Wang et al., 2008). Fig. 4 illustrates the retinal gliosis and outer nuclear disorganization observed on retinal sections cut along a vertical meridian passing through the optic nerve (Fig. 4C, D). These retinal lesions were larger in the upper than in the lower hemisphere (Duboc et al., 2004), as in taurine-depleted animals. Taurine supplementation considerably decreased the development of these retinal lesions in VGB-treated animals, and the decrease in photopic ERG amplitude observed in VGB-treated animals and patients, was less marked when taurine was added to the drinking water supplied to the animals (Jammoul et al., 2010, 2009). These studies on the retinal toxicity of VGB showed that taurine depletion induced not only photoreceptor degeneration, with cones degenerating first, but also RGC loss, RGCs appearing as the primary site of damage. This conclusion was confirmed by our recent re-examination of retinal lesions in mice treated with the Tau-T blocker GES (Gaucher et al., 2012).

As in taurine-depleted animals, the occurrence of the retinal lesions induced by VGB was dependent on exposure to light (Jammoul et al., 2009). This light dependence was first reported by others working on *in vitro* retinal tissues incubated in VGB and exposed for a few hours to intense illuminations (Izumi et al., 2004). This light dependence was suggested in the first report on the retinal toxicity of VGB, because retinal lesions were found only in albino animals (Butler et al., 1987). We confirmed this observation by maintaining VGB-treated animals in the dark and showing the disappearance of retinal lesions (Jammoul et al., 2009). In Fig. 4A–C, rod/bipolar synapses were visualized by staining the photoreceptor synaptic ribbons with bassoon antibody (red), whereas rod bipolar cells were immunolabeled with the Go α antibody (green). In control mice, rod/bipolar cell synapses were found exclusively in the outer plexiform layer (Fig. 5A), whereas additional ectopic synapses were found to have formed in the ONL of VGB-treated mice (Fig. 5B). By contrast, no retinal lesions were observed in VGB-treated animals kept in the dark, because no retinal gliosis or disorganization of the photoreceptor layers was observed. Similarly, the formation of ectopic synapses triggered by VGB treatment at the level of rod/bipolar cell synapses was abolished by keeping of the animals in the dark. This study clearly demonstrated that the retinal toxicity of VGB depends strongly on light stimulation. It suggests that patients treated with VGB should try to limit their exposure to bright light.

Although the light dependence of retinal VGB toxicity was demonstrated for photoreceptor damage but not for RGC loss, this

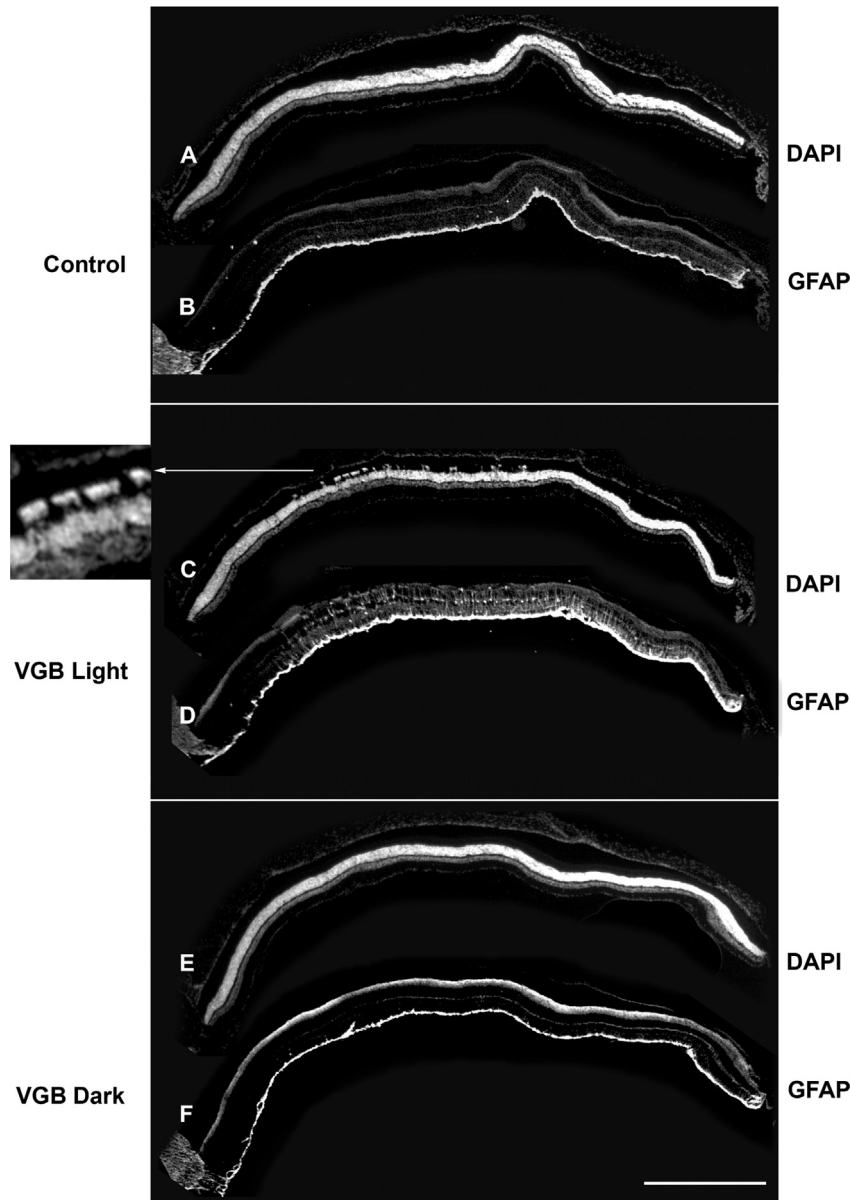


Fig. 4. Light-induced retinal disorganization and retinal gliosis in VGB-treated rats displaying taurine depletion. Whole retinal sections counterstained with the nuclear dye DAPI (A, C, E) and immunolabeled with GFAP antibody (B, D, F). These immunolabelings show a disorganization of retinal layers (B) and retinal gliosis (C) in animals treated with VGB and exposed to light, whereas VGB-treated animals kept in the dark (E, F) display the normal retinal layering (E) and GFAP labeling of glial cells classically observed in control animals (A, B). Note the photoreceptor nuclei displaced above the outer limiting membrane (OLM) (inset in C) and the GFAP-positive processes extending vertically throughout the retina (inset in D) in VGB-treated rats exposed to light. Scale bar: 1 mm.

study also suggested that RGCs might be sensitive to phototoxicity. This conclusion is consistent with the results obtained with cultures of the RGC5 cell line exposed to light (Wood et al., 2007). Blue light appears to be particularly harmful, due to its effects on mitochondrial function (del Olmo-Aguado et al., 2012; Lascaratos et al., 2007; Osborne et al., 2008). This phototoxicity in mitochondria and RGC-5 is counteracted by antioxidants (Costa et al., 2008; Osborne et al., 2008; Wood et al., 2007), suggesting that oxidative stress is a major component of the light-induced RGC death. RGC axons, which contain large numbers of mitochondria, seem to be highly sensitive to phototoxicity, particularly due to the lack of protection of optic fibers by the blue filtering of macular pigments (Osborne, 2008; Osborne et al., 2010). As discussed below, this phototoxicity may be highly relevant in some retinal diseases, such as glaucoma (Osborne, 2008). The findings of these studies on the

phototoxicity of RGCs and its prevention by antioxidants are consistent with phototoxicity in RGCs cells in conditions of taurine depletion and with the neuroprotection of RGCs by taurine.

Following the demonstration of VGB-elicited phototoxicity, we investigated whether VGB administration in the evening (before the night period) could attenuate retinal lesions in patients. Indeed, pharmacokinetic studies have shown that VGB concentration peaks in the hours immediately following intake. Evening administration would therefore result in this peak occurring during the dark period. We compared the retinal lesions occurring in VGB-treated animals receiving their injections in the morning with those occurring in animals treated in the evening (Fig. 6). Strong reactive gliosis (green) and bipolar plasticity (red) were observed in the rats subjected to morning injections of VGB (VGB A.M., Fig. 6B), whereas these effects were weaker in rats receiving their injections later in

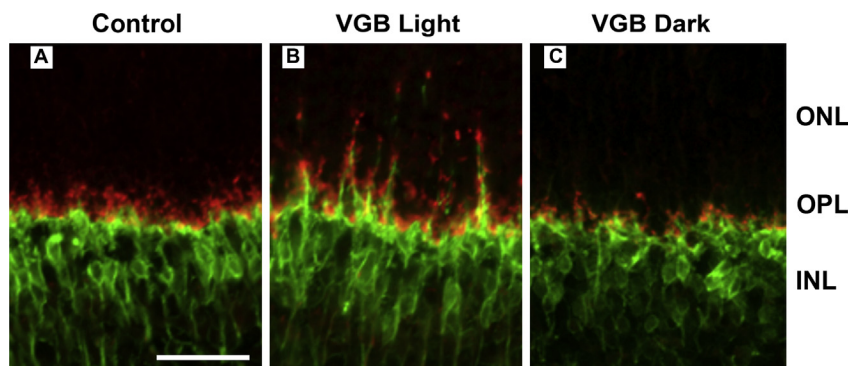


Fig. 5. Light-induced retinal plasticity in VGB-treated mice displaying taurine depletion. Mouse retinal sections immunolabeled with the bassoon (red) and *Gox* (green) antibodies, showing the formation of ectopic photoreceptor/bipolar cell synapses in the ONL of VGB-treated animals exposed to light (B) whereas VGB-treated mice kept in the dark (C) had photoreceptor/bipolar cell synapses in the OPL, as in control animals (A). The bassoon antibody labels synaptic ribbons in photoreceptors, whereas the *Gox* antibody binds to ON bipolar cells. The inset illustrates such ectopic synapses between bassoon-positive structures and growing bipolar cell dendrites. Scale bar: 20 μ m. ONL: outer nuclear layer, OPL: outer plexiform layer.

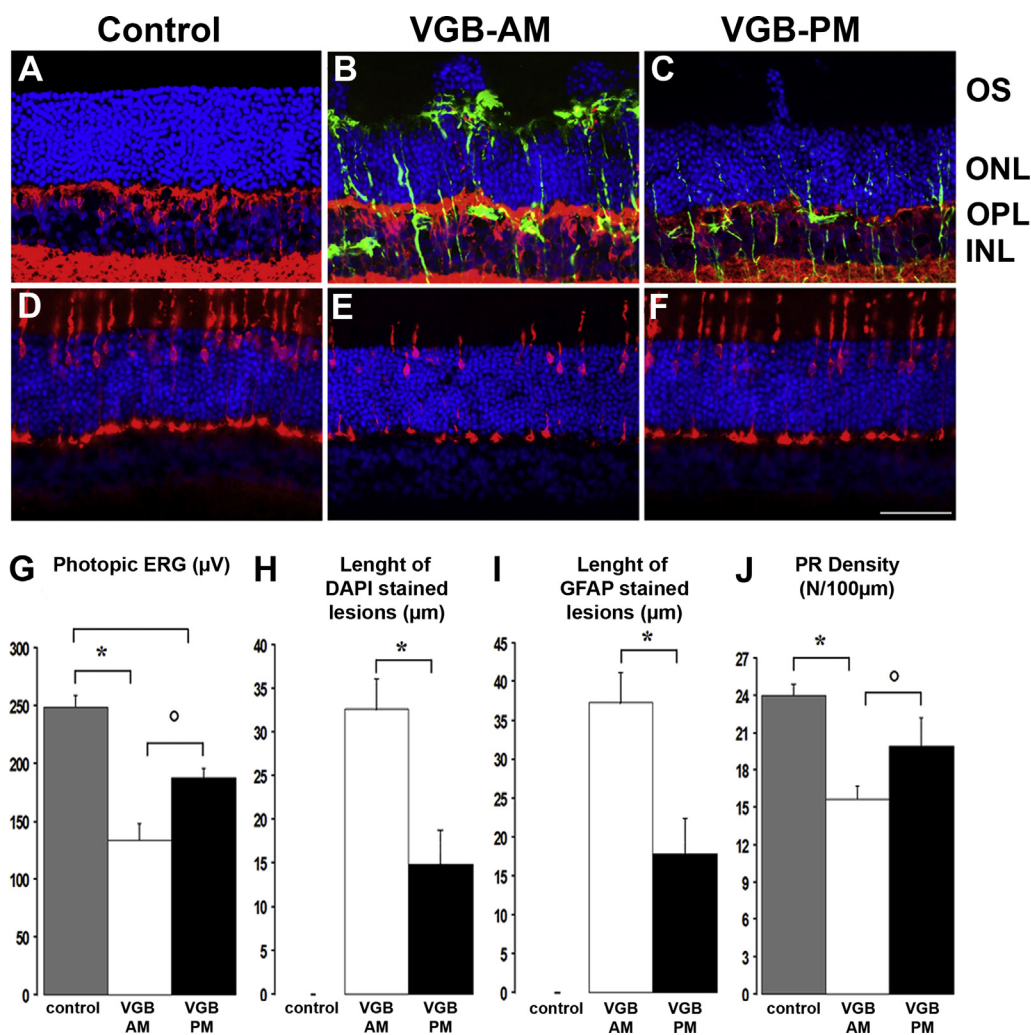


Fig. 6. Time-of-day dependence of VGB-induced phototoxicity. A–F: Retinal sections stained with the nuclear dye DAPI (blue), immunolabeled with GFAP (green; A–C), *Gox* (red, A–C), and cone arrestin (red, D–F) antibodies either in control animals (A, D) or in rats treated with VGB in the morning (VGB-AM; B, E) or in the evening (VGB-PM; C, F); note the intense reactive gliosis (green), ectopic *Gox* staining in ONL (B), and the lower levels of arrestin-positive cone photoreceptors (E) in morning-treated rats (VGB AM), whereas less severe changes are observed in evening-treated rats (VGB-PM; rats C, F). G–J: Quantification of a photopic ERG (G), lengths of displaced photoreceptor nuclei (H), lengths of GFAP-stained areas (I) and cone photoreceptor densities (J) in control animals (gray bar; $n = 9$), morning-treated animals (VGBAM, white bars; $n = 10$), and evening-treated animals (VGBPM, black bars; $n = 10$). The data shown are means $\pm$ SEM from independent animals. * $p < 0.05$; * $p < 0.05$ (One-way ANOVA followed by Bonferroni post hoc test for P, Q, S and Dunn's post hoc test for R). Scale bar = 50 μ m. ONL: outer nuclear layer.

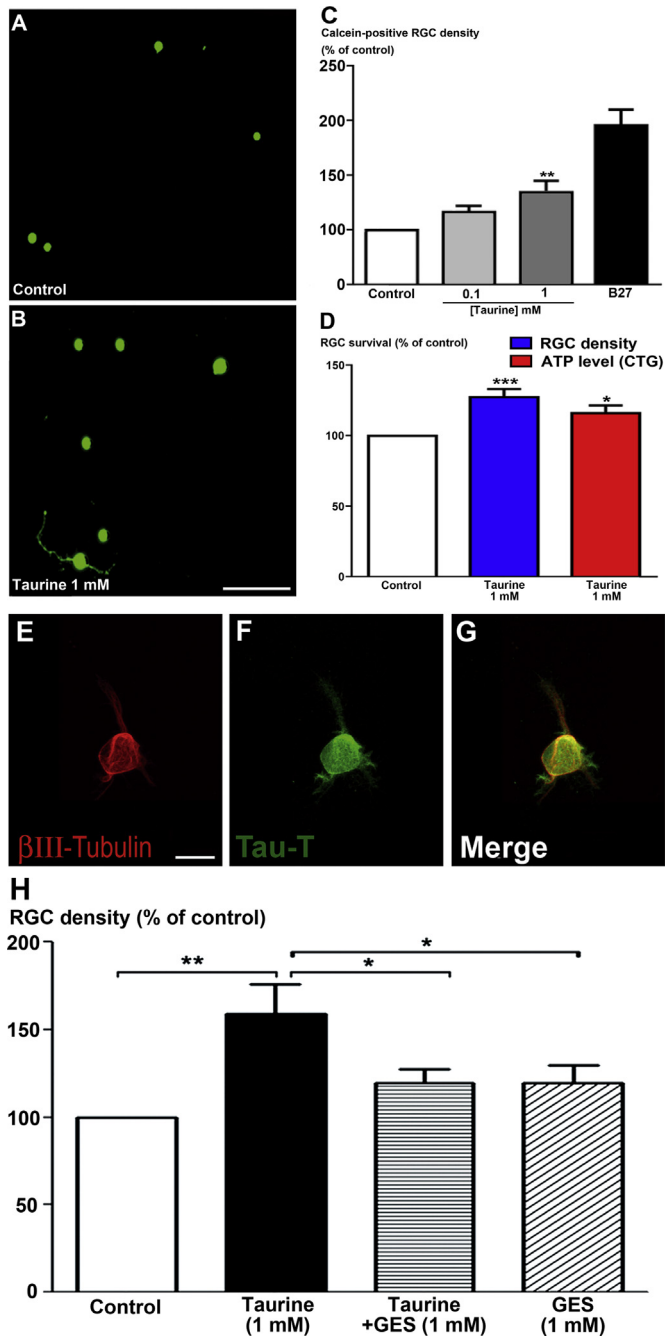


Fig. 7. Taurine transporter expression in retinal ganglion cells (RGCs) and its involvement in the taurine-mediated survival of pure RGCs. A–B: Purified retinal ganglion cells (RGCs) labeled with calcein, visualizing viable cells, after 6 days *in vitro* (DIV). RGCs were cultured in a serum-free medium without (A) or with (B) added taurine (1 mM). The scale bar represents 100 μ m. C: Quantification of viable RGCs at 6 DIV in the presence of two concentrations of taurine (0.1 and 1 mM). In each experiment, viable RGCs were normalized with respect to control levels; the data shown are the means of the experiments carried out (0.1 mM, $n = 8$; 1 mM $n = 21$). Experiments were validated if the addition of B27 supplement taken as positive control. D: Confirmation of the increase in RGC survival mediated by taurine, by ATP measurements. In the same experiments, RGC survival was quantified by both calceinAM labeling, as in A–C, and ATP measurements ($n = 7$). E–G: Confocal images showing taurine transporter (Tau-T) immunolabeling (green; F–G) in β III-tubulin-positive cultured RGCs (red; E, G) at 6 DIV. H: Involvement of the taurine transporter in the taurine-mediated survival of purified RGCs. Quantification of calcein-positive RGCs at 6 DIV, after incubation with 1 mM taurine alone (black bar; $n = 10$), 1 mM taurine plus 1 mM GES (horizontal hatched bar; $n = 9$) or GES alone (oblique hatched bar; $n = 11$). In each experiment, viable RGCs were normalized with respect to control values and the data shown are the means of the various experiments. Note that the enhanced RGC

the day (VGB P.M.; Fig. 6C). In addition, cone density quantification revealed that the morning-treated group had 38% fewer photoreceptors (Fig. 6E) than the control animals receiving saline injections (Fig. 6D), whereas the evening-treated group had 21% fewer photoreceptors (Fig. 6F). Functionally, these cell losses resulted in a 44% decrease in photopic ERG amplitudes in morning-treated rats, versus a decrease of 28% in the evening-treated animals (Fig. 6G). Thus, retinal lesions are clearly smaller following VGB administration just before the dark period than after administration in the morning. Evening administration did not entirely abolish the retinal toxicity of VGB, but it did greatly decrease the extent of the retinal lesions (Fig. 6H–J). As VGB is a non competitive drug blocking GABA-transaminase in an irreversible manner, the efficacy of VGB against epileptic seizures should not be dependent upon the time of administration for treatments requiring a single daily dose of VGB. Our results therefore suggest that patients should be advised to take their VGB treatments in the evening, to minimize the retinal toxicity of VGB.

Our observations (i) of greater retinal lesions in the superior hemisphere, (ii) of the dependence of retinal lesion formation in light conditions and (iii) of the prevention of retinal toxicity by taurine supplementation are different features consistent with the conclusion that the taurine depletion detected by measurements of plasma concentrations is responsible for the retinal lesions. The relevance of this conclusion for VGB-treated patients is provided by the observed low plasma taurine concentrations in these patients. The clinical symptoms and features described in VGB-treated patients can thus provide clinical criteria for the diagnosis of retinal taurine depletion of dietary, genetic, biological or physical origin.

2.2.3. Mechanisms of taurine-induced RGC neuroprotection

2.2.3.1. Effect on purified RGCs. Following the discovery that taurine depletion could cause RGC degeneration, as in the retinal toxicity of vigabatrin, we investigated the mechanisms by which taurine promoted RGC survival (Froger et al., 2012). *In vivo*, it was difficult to determine whether the neuroprotective effect of taurine was mediated by a direct effect on RGCs or whether it required interactions with other cell types. We addressed this question by culturing purified adult rat RGCs in conditions of serum deprivation, thus mimicking ischemic conditions (Fuchs et al., 2005). Under these conditions, the addition of 1 mM of taurine to the culture medium significantly enhanced RGC survival over six days *in vitro* (6 DIV) (Fig. 7A–D). Our freshly purified RGCs expressed Tau-T (Fig. 7E–G) (Froger et al., 2012), as previously reported for the immortalized cell line RGC5 (El-Sherbeny et al., 2004). We therefore investigated the role of this transporter in the observed survival of RGCs *in vitro*. The addition of GES, a competitive blocker of Tau-T, to the culture medium abolished the taurine-mediated survival effect on RGCs (Fig. 7H). These results are consistent with a role for intracellular mechanisms in the neuroprotective effect of taurine on RGCs.

2.2.3.2. Excitotoxicity. We assessed the mechanisms underlying taurine-mediated neuroprotection further, by investigating whether taurine could modulate glutamate excitotoxicity, which is known to trigger a massive calcium influx (Choi, 1987). In other neuronal cell types, taurine has indeed been shown to prevent glutamate excitotoxicity (Chen et al., 2001; El Idrissi and Trenkner,

survival mediated by taurine is abolished by the addition of GES to taurine-containing medium. The data shown are means $\pm$ SEM from independent experiments. ** $p < 0.001$; * $p < 0.01$ and * $p < 0.05$ versus the control. One-way ANOVA followed by a Dunns post-hoc test. The scale bar represents 10 μ m.

1999), by enhancing Ca^{2+} buffering by mitochondria (El Idrissi, 2008; El Idrissi and Trenkner, 2003). We addressed this effect of taurine on RGCs, by exposing *ex vivo* retinal tissue explants to the glutamate receptor agonist NMDA (100 μM) (Froger et al., 2012). RGCs in these retinal explants were identified by immunolabeling with an antibody directed against Brn-3a, as previously reported (Nadal-Nicolas et al., 2009). By following this strategy, we showed that the incubation of retinal explants with NMDA triggered a severe loss of RGCs (Fig. 8A–E). This NMDA-elicited RGC loss was associated with the induction of apoptosis in the GCL (Fig. 7G). The addition of taurine together with NMDA in the culture medium resulted in a significantly smaller number of apoptotic cells in the

GCL (Fig. 8H, I). This co-application of taurine thus decreased the NMDA-elicited loss of RGCs, as demonstrated by immunolabeling for Brn-3a (Fig. 8D–E). These data demonstrate that taurine can prevent glutamate excitotoxicity and the ensuing apoptosis directly in RGCs.

2.2.4. Taurine in retinal diseases with RGC degeneration

2.2.4.1. Glaucoma. Glaucoma is the leading disease involving RGC degeneration and the second most common cause of blindness worldwide and in industrialized countries. The mechanisms underlying RGC loss are not fully understood, although glutamate excitotoxicity and an increase in oxidative stress have been

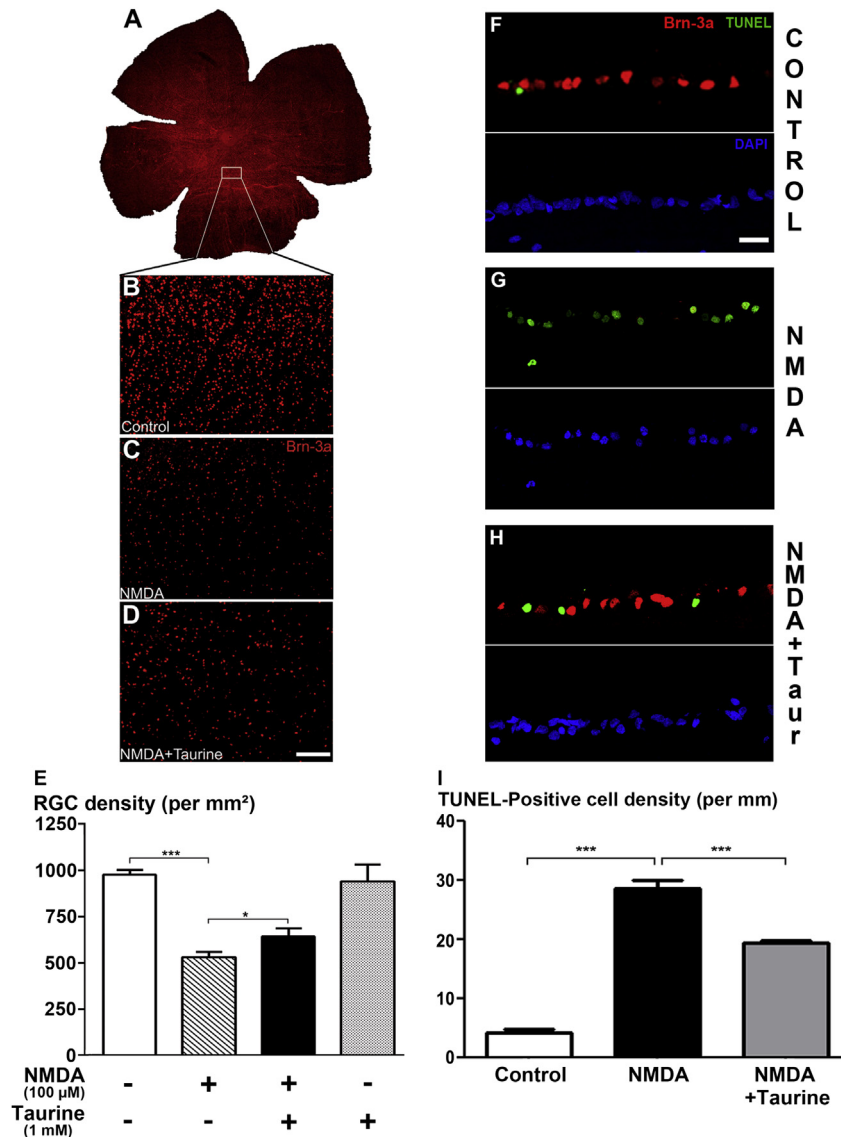


Fig. 8. Taurine-induced prevention of the NMDA-induced apoptosis and loss of retinal ganglion cells (RGCs). A: Digital reconstruction of a flat-mounted retinal explant, immunolabeled with the Brn-3a antibody. B–D: Enlarged images ($\times 10$ objective) from flat-mounted retinal explants showing Brn-3a-immunopositive RGCs in control untreated conditions (Control; B), in an NMDA-treated explant (100 μM , NMDA; C), in an NMDA-treated explant co-incubated with taurine (1 mM taurine, NMDA + Taurine; D). After the culture period (4 DIV), these retinal images were acquired on an automated platform. Scale bar: 100 μm . E: Quantification of RGC densities on flat-mounted retinal explants for control explants ($n = 33$, white bar); NMDA-treated explants ($n = 31$, hatched bar); NMDA- plus taurine-treated explants ($n = 23$, black bar), and taurine-treated explants ($n = 6$, gray bar), respectively. Data are expressed as densities of Brn-3a-immunopositive RGCs (cell/mm²). Cells were counted on all acquired retinal images shown in B–D, in an automatic manner, on the automated platform. F–H: Vertical retinal explant sections showing DAPI-stained nuclei (blue), TUNEL-positive RGCs (green) immunolabeled with the Brn-3a antibody (red) in a control retinal explant (F), in a NMDA-treated retinal explant (G) in a NMDA- plus taurine-treated retinal explant (NMDA + Taur, H). Note that all TUNEL-positive cells in the RGC layer were negative for the RGC marker, Brn-3a. Images of complete retinal explant sections were acquired with a digital fluorescence scanner (Nanozoomer). Scale bar: 20 μm . I: Quantification of TUNEL-positive cells on whole vertical sections as in (F–H) from control retinal explants ($n = 6$, white bar), from NMDA-treated explants ($n = 6$, black bar) from NMDA- plus taurine-treated explants ($n = 6$, gray bar). Data are expressed as TUNEL-positive cell densities per mm of retinal section. Data are means $\pm$ SEM from independent explants. *** $p < 0.001$, * $p < 0.05$ (one-way ANOVA followed by a Bonferroni post-hoc test). Fig. 7A–E were adapted from Froger et al. (2012).

identified as involved in this degenerative process (Seki and Lipton, 2008; Tezel, 2006). Furthermore, it has been suggested that light exposure contributes to the loss of RGCs in the optic neuropathies (Osborne, 2008). Following our demonstration of the importance of taurine for RGC survival and its neuroprotective effect against RGC glutamate excitotoxicity and, possibly, RGC phototoxicity, we hypothesized that taurine might be neuroprotective in glaucoma. For this purpose, taurine supplementations were performed in various animal models of glaucoma, both displaying a significant increase in intra-ocular pressure (IOP): (i) the DBA/2J mouse, which develops an inherited pigmentary glaucoma (Anderson et al., 2002; John et al., 1998) and (ii) rats with epiclinal vein occlusion (Shareef et al., 1995). In both models, the addition of taurine to drinking water doubled plasma taurine concentration with no effect on IOP levels (Froger et al., 2012, 2013). Such increase of plasma taurine concentration, following taurine supplementation through drinking water, was already reported by other studies (Das and Sil, 2012; Li et al., 2013), although in some cases, taurine supplementation did not change the plasma level. In this case, an inhibition of endogenous biosynthesis by the exogenous taurine supply may explain the unchanged taurine levels. In both models, taurine supplementation resulted in higher RGC numbers and this increase in cell survival was associated with a better preservation of retinal cell function, as indicated by ERG recordings. It remains unclear how taurine supplementation mediated the neuroprotection of RGCs in these animal models of glaucoma, in which plasma taurine concentrations were not initially lower than in healthy animals. In fact, the increase in IOP observed in patients with glaucoma and in animal models of this disease is known to limit retinal blood perfusion (Araie et al., 2009). This limitation of blood perfusion probably impairs the retinal uptake of taurine and other micro-nutrients with antioxidant properties (see Fig. 8). These results further demonstrate the importance of taurine for the maintenance of RGCs in normal and pathological conditions.

The absence of photoreceptor damage in glaucoma (Kendell et al., 1995) suggests that taurine depletion may be limited to the inner retina, with no change in the region of photoreceptors in the outer retina. However, studies based on the use of various techniques have also reported photoreceptor degeneration in glaucoma patients (Choi et al., 2011; Panda and Jonas, 1992; Velten et al., 2001). Indeed, very large increases in IOP may also affect choroidal blood flow, thereby causing anatomical changes (Kubota et al., 1996). As a consequence, taurine depletion may also occur in photoreceptors, as observed in dogs with glaucoma, leading to the degeneration of these cells (Madl et al., 2005). Thus, very low levels of vascular blood flow may result in taurine depletion in both the inner and outer retina during glaucoma, thereby contributing to the degeneration of both RGCs and photoreceptors.

2.2.4.2. Retinitis pigmentosa (RP). In RP, photoreceptor loss is followed by a secondary loss of RGCs, with a complete histological reorganization of the retina (Humayun et al., 1999; Marc et al., 2003; Milam et al., 1998). This RGC loss is also observed in various animal models of the disease (Garcia-Ayuso et al., 2010; Kolomiets et al., 2010; Villegas-Perez et al., 1998; Wang et al., 2000). Massive vascular atrophy occurs in parallel to the RGC loss (Froger et al., 2012; Milam et al., 1998; Penn et al., 2000), suggesting a possible relationship between these two events. In RP patients, a decrease in ocular blood flow has been also described in the retina, choroid and retroocular vessels (Cellini et al., 2010; Falsini et al., 2011; Grunwald et al., 1996; Konieczka et al., 2012). The resulting blood vessel attenuation and the formation of circumferential vessels were shown in Royal College of Surgeon rats, an animal model of RP, to cause abrupt changes in the trajectory of RGC axon bundles (Villegas-Perez et al., 1998), leading thereby in their

retrograde degeneration due to blockade of axonal transport. As an alternative hypothesis, we investigated whether the vascular atrophy could decrease taurine retinal uptake, contributing thereby to the RGC loss. We tested this hypothesis, by adding taurine to the drinking water of P23H rats, another rat model of RP, which display RGC loss (Kolomiets et al., 2010) and vascular atrophy (Froger et al., 2012). This taurine supplementation resulted in a larger number of surviving RGCs at 12 months (Froger et al., 2012, 2013), indicating that vascular atrophy may cause retinal taurine levels to decrease below the threshold concentration required for RGC survival (Fig. 8). This conclusion is consistent with the reported decrease in retinal taurine concentration in a dog model of RP (Schmidt and Aguirre, 1985). This increase in RGC survival following taurine supplementation may account for the improvement or stabilization of visual acuity in RP patients taking drug combinations including taurine (Pasantes-Morales et al., 2002).

2.2.4.3. Diabetic retinopathy. Diabetes mellitus (DM) comprises two subtypes of the disease: type I DM caused by insulin deficiency and type II DM resulting from insulin resistance, both leading to hyperglycemia. Diabetic retinopathy is one of the most widespread complications of this disease and is the leading cause of blindness in individuals under the age of 50 years (Aiello et al., 1998; Antonetti et al., 2012; Cai and Boulton, 2002). Its prevalence has been estimated at between 20 and 35% in diabetic patients, resulting in a more than 90 million affected people worldwide (King et al., 1998; Yau et al., 2012). It causes blindness by triggering macular edema and a form of glaucoma (Tumosa, 2008). RGC apoptosis even occurs as an early sign of diabetic retinopathy in patients and animal models (Barber et al., 1998). DM edema results from blood-retinal barrier leakage in the foveal area (Antonetti et al., 2012). The pathological mechanisms which associate the microvascular damage and RGC apoptosis in this condition remain unclear, but seem to involve oxidative stress, as in other complications of DM (Baynes and Thorpe, 1999).

In patients, chronic hyperglycemia is associated with systemic taurine depletion in type I DM (Franconi et al., 1995), type II diabetes (De Luca et al., 2001; Merheb et al., 2007) and in women with gestational diabetes (Seghieri et al., 2007). In patients with type II DM, the presence of diabetic retinopathy has been shown to be specifically associated with a decrease in Tau-T expression in the mononuclear peripheral blood cells whereas these cells have an increase in Tau-T expression in other patients with no retinal complications (Bianchi et al., 2012). One study in men at high risk of type II DM investigated the potential preventive value of taurine, but obtained no positive results (Brons et al., 2004). Another study found no effect of taurine on glycosylated hemoglobin or fasting glucose concentrations (Chauncey et al., 2003), whereas a Russian study (2011) have reported some improvement of metabolic syndrome and diabetes following a combined treatment, including taurine. Taurine was identified as an “anti-diabetic” factor in the 1930s by Ackermann and Heinsen, who were the first to report its hypoglycemic effect (Ackermann and Heinsen, 1935), which was subsequently confirmed (Cherif et al., 1996; De la Puerta et al., 2010; Kulakowski and Maturo, 1984; L'Amoreaux et al., 2010; Maturo and Kulakowski, 1987). Indeed, taurine supplementation can prevent hyperglycemia and protect pancreatic cells during the induction of diabetes by streptozotocin (Alvarado-Vasquez et al., 2003; Tokunaga et al., 1979, 1983) or alloxan (Das and Sil, 2012; Tenner et al., 2003; Winiarska et al., 2009). In a genetic rat model of type II DM, the Otsuka Long-Evans Tokushima Fatty rat (OLETF), taurine supplementation improves both hyperglycemia and insulin resistance (Harada et al., 2004; Kim et al., 2012; Nakaya et al., 2000) and thus increases the survival of these diabetic rats (Di Leo et al., 2004).

Taurine was found to increase the insulin-mediated glucose uptake in heart or muscle tissues (Haber et al., 2003; Lampson et al., 1983). Although some have attributed this taurine regulation of glucose homeostasis to the direct activation of insulin receptors (Maturo and Kulakowski, 1988), it is more likely affecting insulin release from pancreatic beta cells (L'Amoreaux et al., 2010). If this effect could be due to taurine action on voltage-sensitive calcium channels, it could also result from taurine activation of GABA receptors (Braun et al., 2010; Cuttitta et al., 2013; El Idrissi et al., 2009; Kawai and Unger, 1983; Satin and Kinard, 1998). Conversely, hyperglycemia reduces Tau-T expression and activity in both diabetic patients and animal models (Askwith et al., 2009; Stevens et al., 1999), leading to taurine depletion from the plasma (Franconi et al., 1996; Trachtman et al., 1995) and various tissues and organs, including the sciatic nerves (Stevens et al., 1999), lens (Malone et al., 1993) and kidneys (Trachtman et al., 1993).

In diabetic animals, taurine supplementation has been shown to prevent most of the complications of DM, through its cellular and molecular effects on glucose metabolism and its antioxidant properties (Ito et al., 2012; Schaffer et al., 2009). It reduces kidney injury (Das and Sil, 2012), improves cardiac and vascular functions (Das et al., 2012; Moloney et al., 2010; Tappia et al., 2011) and decreases serum triglyceride and cholesterol concentrations (Kim et al., 2012; Mochizuki et al., 1999). Similarly, in diabetic retinopathy, it has been suggested that taurine supplementation slows the effects of pathological processes in various animal models (Di Leo et al., 2003; Di

Leo et al., 2004). Taurine has been shown to attenuate retinal reactive gliosis (Zeng et al., 2009) and retinal glial apoptosis (Zeng et al., 2010) in diabetic rats, suggesting both an anti-inflammatory and an anti-apoptotic action. It has also been suggested that taurine prevents glutamate excitotoxicity by increasing glutamate transporter expression, thereby decreasing glutamate levels (Yu et al., 2008). Taurine has also been reported to modulate the overproduction of vascular endothelial growth factor (VEGF), leading to retinal edema and *de novo* angiogenesis in diabetic retinopathy (Obrosova et al., 2001). Our observations that RGC are highly dependent on taurine for survival are consistent with the notion that the taurine depletion reported in diabetic patients (Franconi et al., 1995) (De Luca et al., 2001; Merheb et al., 2007) may be responsible for glaucomatous optic neuropathy in these patients. Surprisingly, no study has investigated the correlation between taurine levels and RGC apoptosis in diabetic animals. This absence of data make reflect the difficulties faced when trying to observe RGC apoptosis reliably in animal models of diabetes (Asnaghi et al., 2003; Barber et al., 1998; Gaucher et al., 2007; Martin et al., 2004; Park et al., 2003).

3. Conclusions

The requirement of taurine for photoreceptor survival was clearly established in the original study on cats fed on casein in the 1970s (Hayes et al., 1975), but the dependence of RGCs on taurine for survival was demonstrated only recently (Froger et al., 2012;

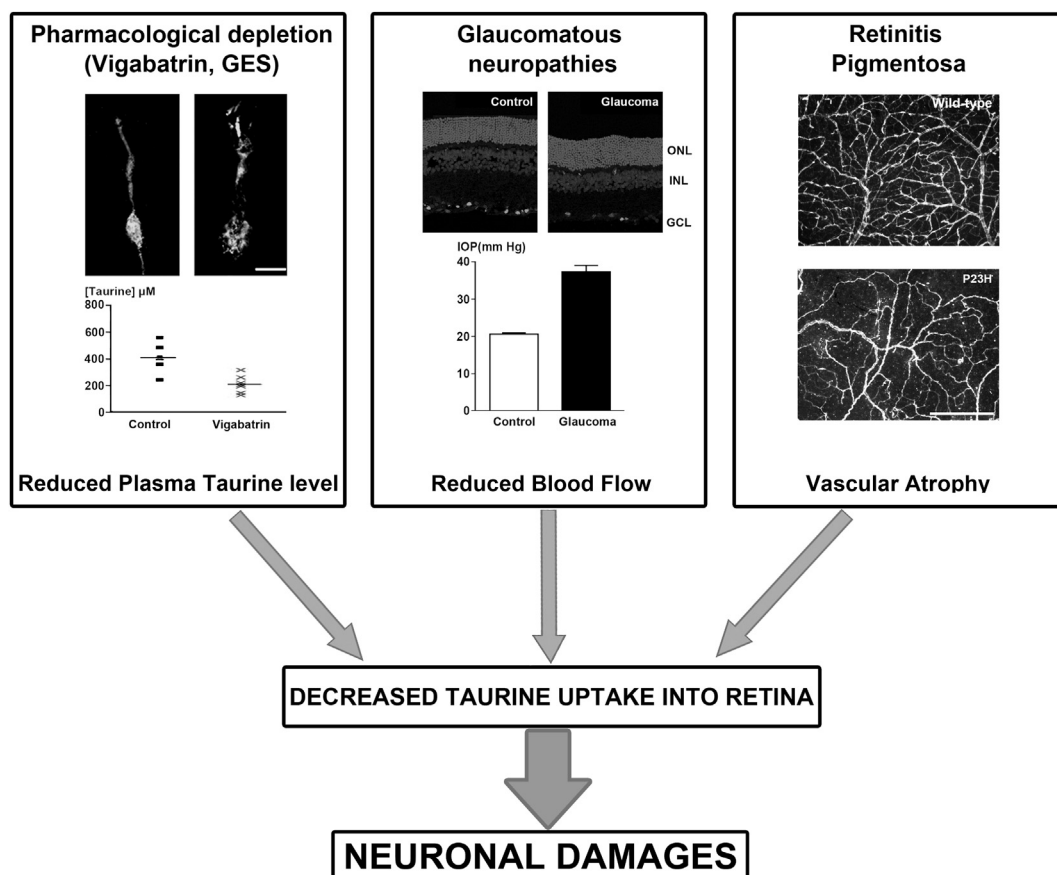


Fig. 9. A unifying taurine hypothesis in various retinal diseases with RGC degeneration. It is proposed here that local retinal taurine deficiency induces RGC degeneration in different retinal diseases as a consequence of different retinal blood flow changes. The retinal taurine deficiency may result from a decrease in plasma taurine concentration, as in VGB toxicity, leading to both cone and RGC degeneration. Similar decreases in plasma taurine concentrations have also been reported in diabetic patients, suggesting a potential role in the development of diabetic retinopathy. Alternatively, the reported decrease in retinal blood perfusion in glaucoma may limit taurine uptake in poorly perfused retinal areas. Finally, the vascular atrophy observed in retinitis pigmentosa limits the surface area available for taurine exchanges between the endothelial cells of the capillaries and the retinal tissue. In all cases, taurine supplementation increases plasma taurine concentrations, at least partly compensating for pathological changes in retinal taurine uptake.

Gaucher et al., 2012; Jammoul et al., 2010). Although initial observations of RGC loss in taurine-depleted animals were occasionally reported prior to these recent studies (Lake et al., 1988; Imaki et al., 1998), this RGC loss was considered a secondary effect of the photoreceptor degeneration whereas we demonstrated a direct taurine dependence for RGC survival (Froger et al., 2012). The observation of RGC loss in VGB-treated patients (Buncic et al., 2004; Frisen and Malmgren, 2003; Ravindran et al., 2001; Wild et al., 2006), also due to taurine deficiency (Jammoul et al., 2009), confers a clinical relevance on this finding that taurine is required for RGC survival. Taurine deficiency must therefore be considered not only in retinal diseases with photoreceptor degeneration, but also in retinal diseases with RGC loss. In such diseases with RGC loss, taurine deficiency may occur due to a decrease in plasma taurine levels and blood components, as in the retinal toxicity of VGB (Jammoul et al., 2009) and in diabetes (Bianchi et al., 2012; De Luca et al., 2001; Franconi et al., 1995; Merheb et al., 2007; Seghieri et al., 2007). However, taurine depletion in the retina may also result from decreases in retinal blood perfusion due to increases in IOP, as in glaucoma (Araie et al., 2009), or vascular atrophy, as in late stages of retinitis pigmentosa (Fig. 9) (Milam et al., 1998). Poor retinal blood perfusion would limit taurine uptake, because blood flow may cease or be severely slow in small capillaries, whereas vascular atrophy would reduce the surface area for taurine uptake by capillary endothelial cells (Tomi et al., 2008). These results suggest that taurine depletion, whether systemic (plasma) or local (retinal) should be considered whenever RGC degeneration is detected in pathological conditions. A few clinical studies have already investigated the possible effects of taurine in the treatment of diabetic retinopathy, but more are required to determine whether this compound can interfere with retinal degenerative processes involving the loss of cone photoreceptors or RGSs.

Taurine has a direct protective effect on isolated RGCs (Froger et al., 2012, 2013). This neuroprotective effect has been attributed to intracellular mechanisms, because RGCs express Tau-T, and Tau-T blockers have been shown to abolish this effect. However, despite numerous studies on taurine neuroprotection in photoreceptors and other neurons over the last 40 years, the molecular mechanisms underlying this neuroprotective effect remain unclear. Further studies are therefore required to investigate, at the cellular level, the way in which taurine protects cells against oxidative stress and excitotoxicity and to determine whether other drugs can activate these neuroprotective pathways.

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